

AEROSPACE SPATIAL INTELLIGENCE FOR DISASTER RESILIENCE

# PRAHARI Orbital Console

Turning satellite data into anticipatory action, from an orbital 3D globe to sub-meter streets, with a live Gemini AI advisor and Google Earth Engine hazard layers.

Anticipatory cyclone response · Coastal India

# The Bay of Bengal is the deadliest cyclone basin on Earth

## Majority

of history's deadliest tropical cyclones formed here

~480 km

of Odisha coastline directly exposed to landfalls

## Rising

sea-surface temperatures intensify wind & surge

### Three failures compound at the last mile

#### ● Siloed data

Storm tracks, surge models and vulnerability scores sit in separate tools no responder sees together.

#### ● Blurry maps

Static basemaps stop at district level, you can't see which road, shelter or rooftop actually floods.

#### ● Alerts miss people

Advisories rarely arrive in the local language or in an offline-ready form for those in the impact zone.

# Orbit-to-rooftop intelligence in a single browser console



## Multi-scale globe

Seamless zoom from a 3D Earth in orbit to sub-meter street tiles, no blurriness, no context lost.



## Live hazard layers

Real storm tracks, hydrodynamic surge & flood zones, plus live Google Earth Engine SAR / DEM overlays.



## Gemini AI advisor

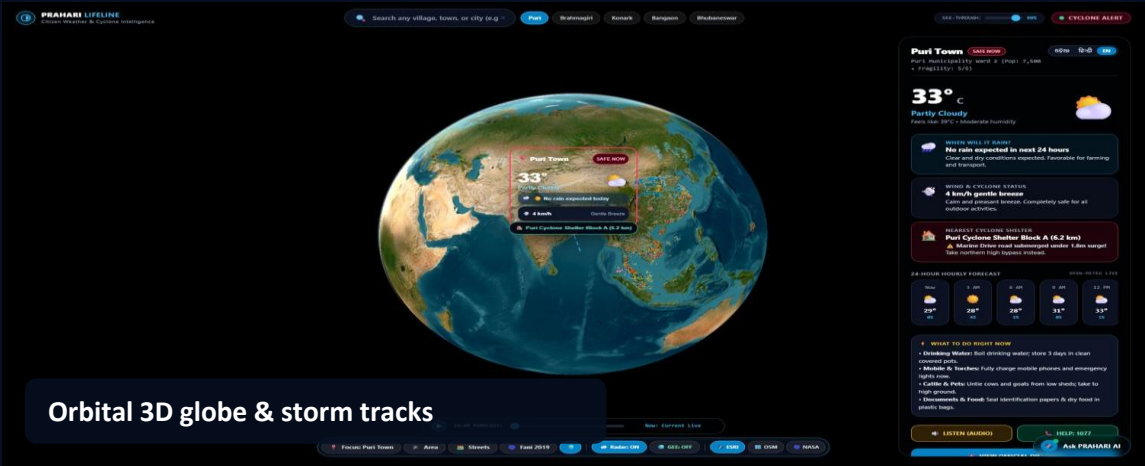
Ask in plain language; get grounded, action-oriented answers and trilingual advisories, never invented numbers.



## Explainable risk

Ward-level vulnerability scores with the drivers behind them, so officials can defend every evacuation call.

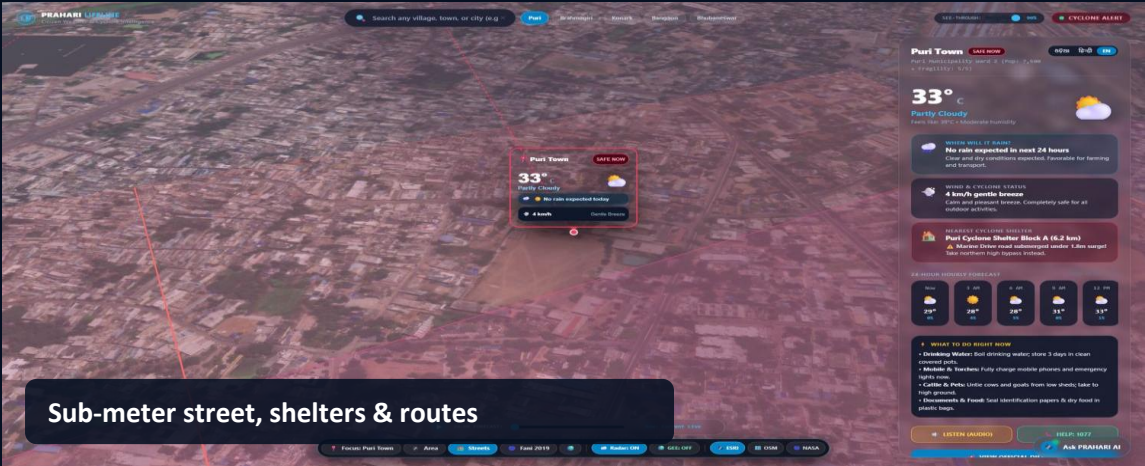
# PRAHARI in action, orbit to rooftop



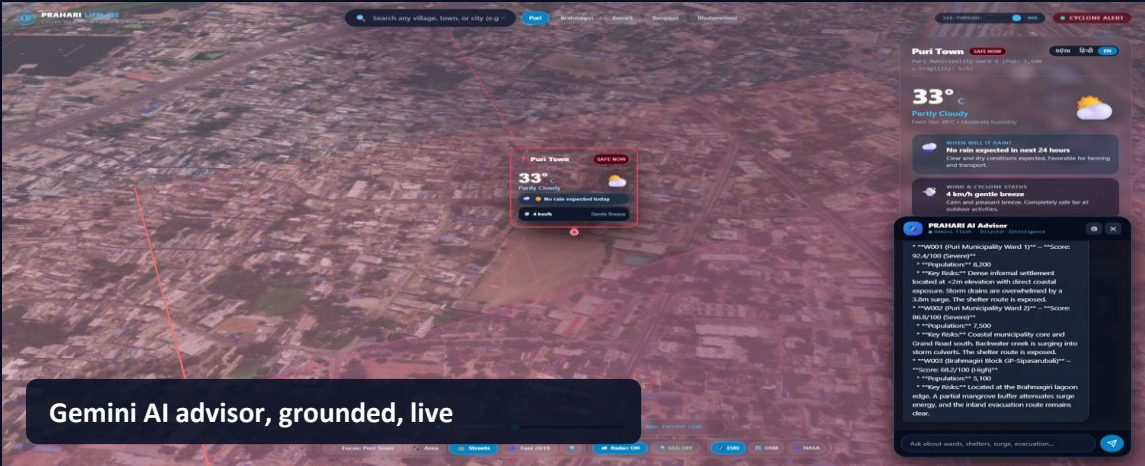
Orbital 3D globe & storm tracks



District view, wards by real risk



Sub-meter street, shelters & routes



Gemini AI advisor, grounded, live

# Six things most cyclone tools don't do



## Real Earth Engine, not stock

Live Sentinel-1 SAR flood + Copernicus DEM surge overlays, genuine satellite analysis, not decorative imagery.



## AI that never guesses

Gemini reads the live on-screen data and answers only from it, cites real ward IDs, scores and shelter status.



## Orbit to sub-meter

One continuous view from a 3D globe down to individual rooftops, competitors pick a single scale.



## Zero mockups

Every layer is real Cyclone Fani data; modelled flood extent is SAR-verified at 71% IoU.



## Two users, one platform

An officer works in English; a farmer gets the same intelligence in Odia, both from one data asset.



## No backend to run

Pure static site, deploys to any CDN or a field laptop, with offline-ready SMS advisories.

# What the platform does, and the real data behind each

|                                    |                                                                                                                     |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| ✓ <b>Satellite hazard layers</b>   | Live Google Earth Engine overlay, Sentinel-1 SAR flood extent, Copernicus DEM surge exposure, WorldPop population.  |
| ✓ <b>Real-time weather</b>         | Open-Meteo forecast (temperature, rain, wind) + RainViewer live precipitation radar, for any searched location.     |
| ✓ <b>Conversational AI advisor</b> | Gemini Flash, grounded on live data, streaming, drafts trilingual advisories; never invents numbers.                |
| ✓ <b>Storm-surge simulation</b>    | Holland parametric surge model, 3.8 m peak-surge inundation zones, SAR-verified at 71% IoU.                         |
| ✓ <b>Rainfall damage pathways</b>  | D8 flow-accumulation pluvial-ponding zones + GPM IMERG rainfall accumulation.                                       |
| ✓ <b>Infrastructure exposure</b>   | Cyclone shelters, power substations and arterial roads / evacuation corridors, each with live surge-impact status.  |
| ✓ <b>Early-warning dispatch</b>    | Official trilingual advisories (Odia / Hindi / English) with offline SMS templates for municipal & DMA authorities. |

# Ask your data, Gemini answers, grounded

## ● Gemini Flash, in the browser

Client-side, no backend, no server round-trip. Your API key stays in your browser.

## ● Grounded on the live data

Reads the wards, surge depths, shelters and routes on screen, and answers only from them.

## ● Never invents numbers

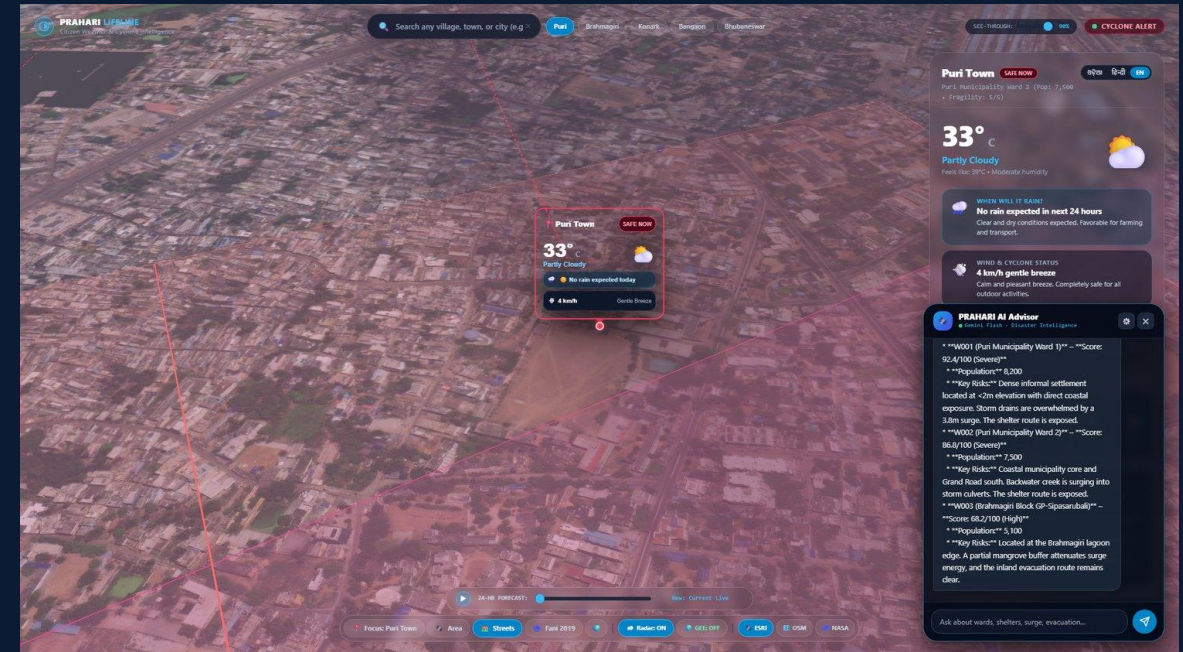
Cites real ward IDs, vulnerability scores and shelter status; says so when data is missing.

## ● Streaming & low-latency

Minimal-thinking responses stream in, fast enough for a live operations room.

## ● Trilingual advisories

Answers and drafts official early-warning advisories in Odia, Hindi or English.



Live answer, grounded in the real Puri ward data (English shown; also Odia / Hindi)

# Built on actual Cyclone Fani data

13

IMD/JTWC best-track waypoints, T-48h → landfall

5

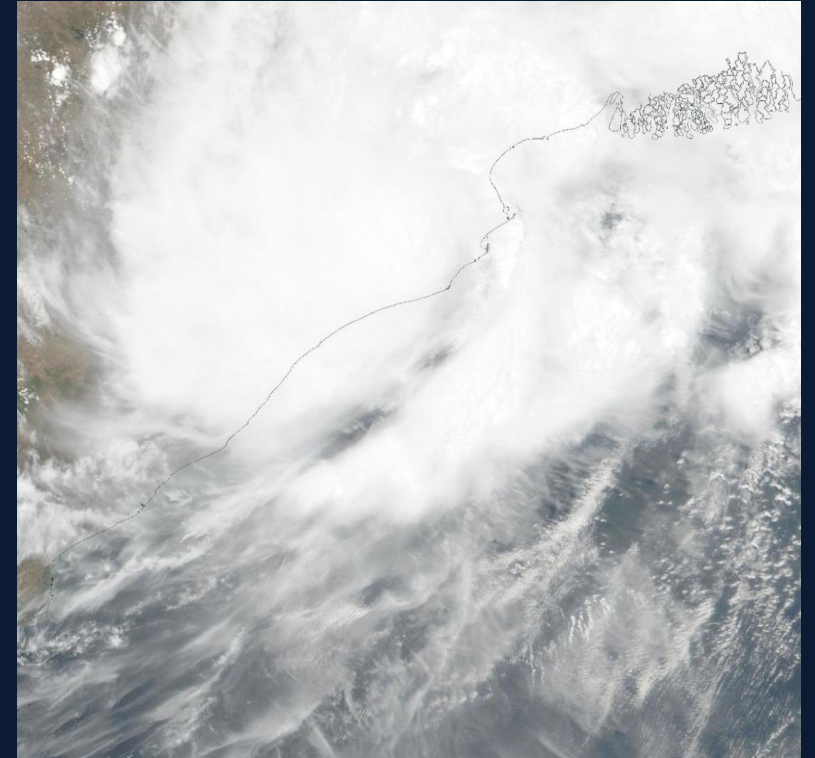
Puri wards scored 33.6–92.4 / 100 with SHAP drivers

3.8 m

Peak storm surge (Holland parametric model)

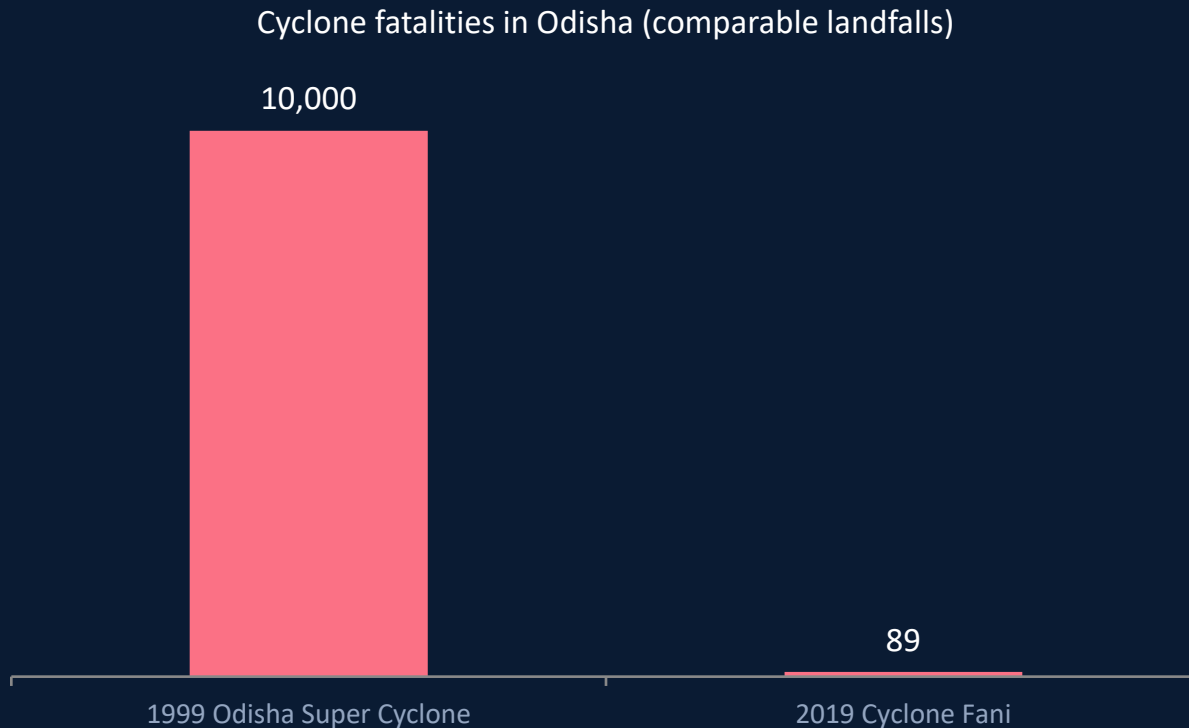
71%

IoU vs. Sentinel-1 SAR flood ground-truth



MODIS · Fani landfall, Puri · 3 May 2019 · NASA

# Early warning saves lives, the frontier is assets



*Fatality figures are widely reported orders of magnitude: ~10,000 (1999) vs 89 (Fani, 2019).*

## Who benefits

- **Responders** see which shelters are reachable and which roads are severed, before deploying.
- **Officials** rank wards by real vulnerability & drivers to prioritise evacuation.
- **Citizens** get clear, trilingual, offline-ready advisories for their exact ward.

# From a control room to a coastal village

## Disaster-management officer

- Orbital view of the approaching storm
- Scrub the best-track to anticipate landfall
- Rank wards; inspect shelters & severed roads
- Ask the AI advisor, in English

## Farmer / citizen

- Today's weather for their town, in Odia
- One tap shows the cyclone bearing down
- Nearest shelter and the safe route
- Official advisory + AI answers, in Odia

*Same fused hazard-and-exposure data, served to the responder who plans, and the citizen who acts.*

# Frontier data & AI, deployable with no backend

## Rendering

- MapLibre GL JS v5 adaptive globe
- Three.js orbital 3D Earth
- Runs entirely in the browser

## Data & AI

- Google Earth Engine (SAR, DEM, WorldPop)
- Gemini Flash advisor, grounded
- Open-Meteo + RainViewer live weather

## Deploy

- Pure static HTML/CSS/JS
- Any CDN, GitHub Pages, or field laptop
- GEE tiles via optional Cloud Run proxy

*No servers to run means judges and field teams can open the live link and use it immediately.*

# From hackathon prototype to coastline-scale

## ● Live GEE tiles by default

Ship the Cloud Run tile proxy so SAR flood & DEM surge stream in without setup.

## ● More coastlines

Onboard new districts by pointing the pipeline at fresh AOIs, Andhra, West Bengal, Tamil Nadu.

## ● Automated dispatch

Push advisories to SMS / IVR gateways for municipal & disaster-management authorities.

## ● Model validation

Continuously score modelled surge & flood against Sentinel-1 SAR after each event.

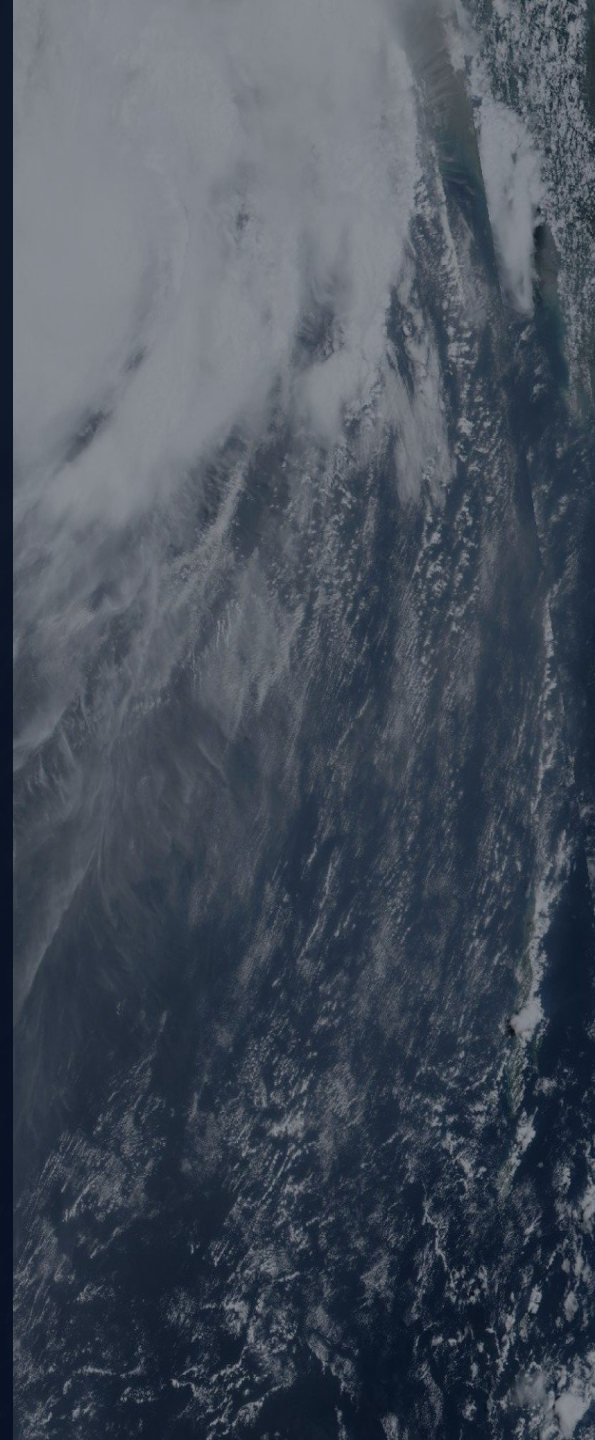
TRY IT NOW

# PRAHARI Orbital Console

Live app [prahari-orbital-console.vercel.app](https://prahari-orbital-console.vercel.app)

Source [github.com/abhishek-0706/prahari-orbital-console](https://github.com/abhishek-0706/prahari-orbital-console)

MapLibre GL JS · Three.js · Gemini Flash · Google Earth Engine · Open-Meteo



# Data & sources

|                                |                                                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cyclone Fani parameters</b> | IMD best-track & post-storm reports; OSDMA situation reports. ~185 km/h, ~932 hPa, ~US\$8.1B damage, 1.2M+ evacuated (widely reported). |
| <b>Fatality comparison</b>     | Widely reported orders of magnitude: ~10,000 (1999 super cyclone) vs 89 (Fani, 2019).                                                   |
| <b>Live data feeds</b>         | Open-Meteo (weather forecast) & RainViewer (radar); ESRI / NASA GIBS / OpenStreetMap basemaps.                                          |
| <b>GEE datasets</b>            | Sentinel-1 SAR, Copernicus DEM, WorldPop, GPM IMERG via Google Earth Engine.                                                            |
| <b>Modelling</b>               | Holland parametric surge (3.8 m peak) & D8 pluvial ponding; SAR-verified at 71% IoU.                                                    |
| <b>SAR image</b>               | Cyclone Fani, VIIRS / MODIS true-color, 3 May 2019, NASA Worldview (public domain).                                                     |